

Comparison of Predictive Models for Insurance Claim Frequency

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Data Preparation

- The sample size was reduced to 200,000 observations from the `freMTPL2freq` dataset.
- The dataset was divided into three parts:
 - **Training set (70%):** Hyperparameter optimization of algorithms using 5-fold cross-validation (K-fold CV).
 - **Validation set (15%):** Optimization of the number of iterations (M) via the *early stopping* mechanism.
 - **Estimation (Train + Valid):** Final models were trained on the combined set (85% of the data).
 - **Test set (15%):** *Out-of-Sample* error evaluation.

Feature engineering

- Transformations (e.g., logarithm for Density) and discretization of continuous variables (`VehAge`, `DrivAge`) were applied.

Characteristics of baseline models

1. Null Model:

- The simplest estimator, not accounting for any explanatory variables (estimation of the intercept β_0 only).
- The model equation for the expected value $\mu_i = \mathbb{E}[\text{ClaimNb}_i]$:

$$\ln(\mu_i) = \beta_0 + \ln(\text{Exposure}_i) \implies \mu_i = \text{Exposure}_i \cdot \exp(\beta_0)$$

2. Generalized Linear Model (GLM):

- Model equation:

$$\ln(\mu_i) = X_i\beta + \ln(\text{Exposure}_i) \implies \mu_i = \text{Exposure}_i \cdot \exp(X_i\beta)$$

Random Forest with Parametric Bootstrap

- Instead of drawing observations, a Parametric Bootstrap was implemented. The new target variable in each iteration is drawn from the theoretical Poisson distribution based on the GLM prediction:

$$Y_i \sim \mathcal{Poisson}(\hat{\lambda}_{GLM} \cdot Exposure)$$

- To reduce correlation among trees in the forest, each node is optimized based on a randomly drawn subset of explanatory variables.
- Five-fold cross-validation was used to select the optimal hyperparameters. The table below presents the hyperparameter values considered.

Table: Hyperparameter optimization grid

Hyperparameter	Search space
Maximum depth (maxdepth)	{5, 10}
Minimum obs. in a leaf (minbucket)	{10, 50}
Number of sampled features (m_features)	{5, 7}

Response Boosting

- In each step m , a weak tree is built, determining a multiplicative in-sample correction $\hat{\psi}^{(m)}(x)$ of the current predictor.
- Nodes are optimized by minimizing the Poisson deviance for observations with an updated exposure $v_i^{(m)}$ and a new target variable $y_i^{(m)}$:

$$v_i^{(m)} = v_i \hat{\mu}^{boost, m-1}(x_i), \quad y_i^{(m)} = \frac{y_i}{\hat{\mu}^{boost, m-1}(x_i)}$$

- The newly returned predictor incorporates the modification scaled by the learning rate α :

$$\hat{\eta}^{boost, m}(x) = \hat{\eta}^{boost, m-1}(x) + \alpha \log \left(\hat{\psi}^{(m)}(x) \right)$$

Response Boosting – Implementation

- The model building process was divided into 3 stages:
 - ➊ **5-fold Cross-Validation (CV):** Selection of optimal algorithm hyperparameters.
 - ➋ **Early Stopping:** Monitoring error on the validation set to find the optimal number of iterations (M).
 - ➌ **Final estimation:** Building the final model on the training and validation data for the determined M .

Table: Hyperparameter optimization grid

Hyperparameter	Search space
Maximum depth (maxdepth)	$\{2, 4\}$
Minimum obs. in a leaf (minbucket)	$\{10, 50\}$
Learning rate (alpha)	$\{0.05, 0.1\}$

Gradient Boosting for Poisson Distribution

- In each step, we build a weak decision tree that corrects the errors of the current model by moving in the direction of the steepest descent of the loss function (negative gradient).
- Due to the logarithmic link function of the Poisson distribution ($\eta = \log \mu$), both algorithms – Response and Gradient Boosting – update the expected value μ multiplicatively.
- The difference lies in the way the error is defined for the tree:
 - **Response Boosting:** The target variable for the tree is the ratio $y_i/\hat{\mu}_i$, accounting for the updated exposure.
 - **Gradient Boosting:** The tree is trained on the negative gradient of the loss function. For the Poisson distribution, the gradient is $R(\hat{\mu}_i, y_i) = 2(y_i - \hat{\mu}_i)$.
- Hyperparameter optimization (CV) and the search for the optimal number of trees M via Early Stopping proceeded analogously to the Response Boosting algorithm.

- The algorithm provides an alternative to gradient boosting by replacing the gradient descent method with a method based on the second-order Taylor expansion for the loss function.
- In the m -th step, local gradient (R) and Hessian (H) values are determined with respect to the predictor η .
- A regularization function $\frac{1}{2}\lambda||\hat{\mu}_t||^2$ is introduced, which constrains the size of the weights in the tree leaves, thereby reducing overfitting.
- The optimal update value for a given leaf t is determined by the formula:

$$\hat{\mu}_t^{(m,*)} = -\frac{\sum_{x_i \in \mathcal{X}_t^{(m)}} v_i R_i^{m-1}}{\sum_{x_i \in \mathcal{X}_t^{(m)}} v_i H_i^{m-1} + \frac{1}{2}\lambda}$$

where v_i denotes the exposure treated as an observation weight.

- The new predictor is defined as:

$$\hat{\eta}^{boost,m}(x) = \hat{\eta}^{boost,m-1}(x) + \alpha \sum_{t \in \mathcal{T}^{(m)}} \hat{\mu}_t^{(m,*)} 1\{x_i \in \mathcal{X}_t^{(m)}\}$$

XGBoost Implementation and Hyperparameter Optimization

- Hyperparameters were tuned using 5-fold cross-validation. The optimal number of iterations M was determined by an *early stopping* mechanism on the validation set, and the final estimator was trained on the combined sample of 85% of the data.

Table: XGBoost hyperparameter optimization grid

Hyperparameter	Search space
Maximum depth (maxdepth)	{2, 4}
L2 regularization parameter (λ)	{1, 10}
Learning rate (α)	{0.05, 0.1}

Comparison of Model Deviances

- An **Ensemble** estimator was defined as the arithmetic mean of predictions from the random forest and both boosting methods.
- The final evaluation was conducted on the test set.

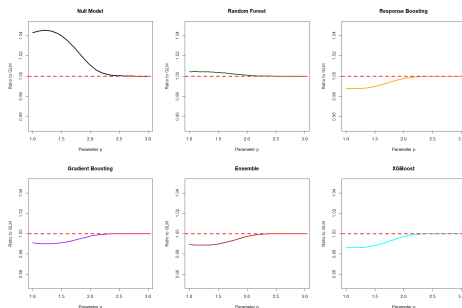
Table: Comparison of Poisson deviances for the evaluated models

Model	Poisson Deviance ($\times 10^2$)
XGBoost	24.17845
Response Boosting	24.20949
Ensemble	24.24372
Gradient Boosting	24.28069
GLM	24.50689
Random Forest (Parametric)	24.59825
Null Model	25.55578

Forecast Dominance Analysis

Tweedie deviance ratio for $p \in [1, 3]$

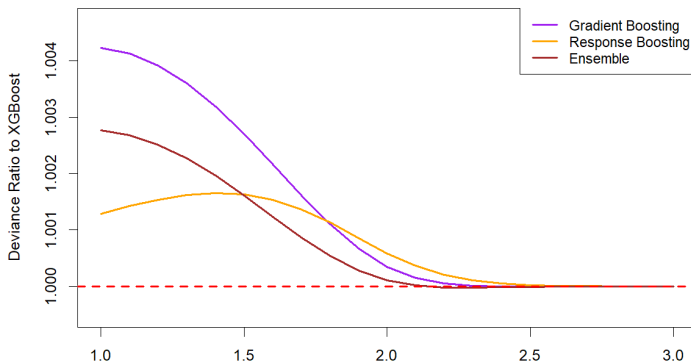
- Predictions of the null model and the random forest are dominated by the reference model (GLM) for every p .
- The Response Boosting, Gradient Boosting, Ensemble, and XGBoost models dominate the remaining models.



Forecast Dominance Analysis

- The chart presents the ratio of deviances of individual models to the **XGBoost** model (reference model) depending on the Tweedie distribution parameter p .
- Across the entire range of the parameter $p \in [1, 3]$, the XGBoost model achieves the lowest deviance values.

Forecast Dominance - Leading Algorithms



Fulfillment of the Global Balance Principle

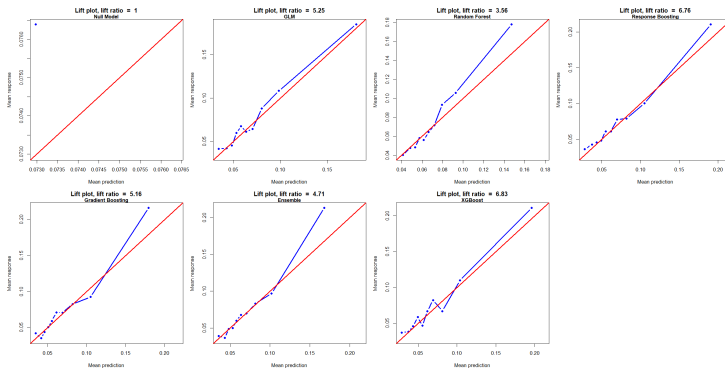
- Consistency of the total expected number of claims with the actual observed number of claims in the portfolio.
- The observed claim frequency in the test portfolio is **7.64%**.

Model	Expected frequency (%)	Deviation (p.p.)
GLM	7.30	-0.34
Null Model	7.30	-0.34
XGBoost	7.26	-0.38
Gradient Boosting	7.25	-0.39
Ensemble	7.23	-0.41
Response Boosting	7.23	-0.41
Random Forest	7.21	-0.43

- Algorithms exhibit a tendency to systematically underestimate the global risk on the test set by approximately 0.3 – 0.4 percentage points. The GLM and Null Model are closest to the actual value.

Local Balance Principle

- The portfolio was divided into 10 deciles according to increasing prediction, to juxtapose the predicted risk with the observed claim frequency in each of them; the *Lift Ratio* measure (the ratio of risk in the highest decile to the lowest) evaluates the power of discrimination of extreme profiles, and the highest result was obtained for the **XGBoost (6.83)** model.



Summary of Model Metrics

- **Calibration error:** Systematic calibration error.
- **Discrimination:** Ability to differentiate risk.
- **ABC:** Area between the concentration curve and the Lorenz curve.
- **Gini Index:** Ability to separate risk before and after model calibration.

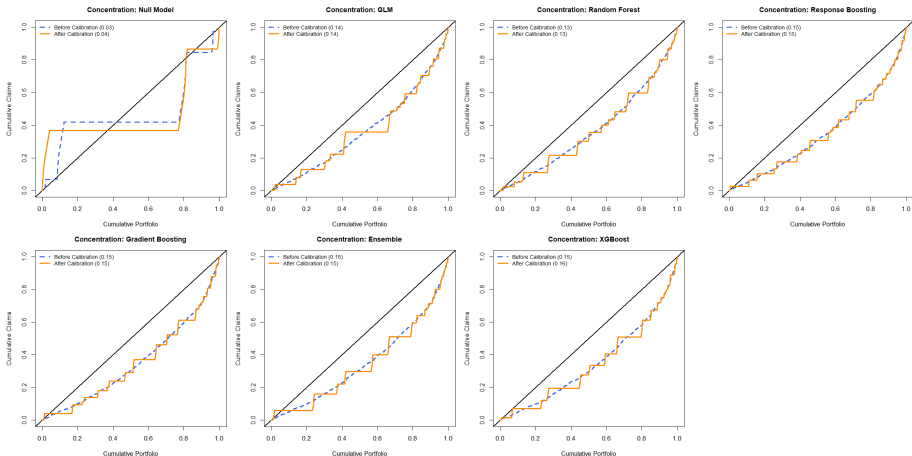
Model characteristics and Murphy's decomposition

Model	Deviance	Calibr. Error	Discrimination	ABC	Gini (Orig.)	Gini (Calib.)
Null Model	25.55	0.136	0.127	0.025	0.052	0.076
GLM	24.51	0.160	1.201	-0.003	0.274	0.284
Random Forest	24.60	0.177	1.127	0.026	0.269	0.280
Resp. Boost.	24.21	0.169	1.507	-0.007	0.309	0.317
Grad. Boost.	24.28	0.198	1.465	0.008	0.309	0.317
Ensemble	24.24	0.193	1.496	0.020	0.310	0.320
XGBoost	24.18	0.157	1.526	-0.006	0.315	0.325

- The **XGBoost** model performs best across key metrics (Deviance, Discrimination, Gini index).

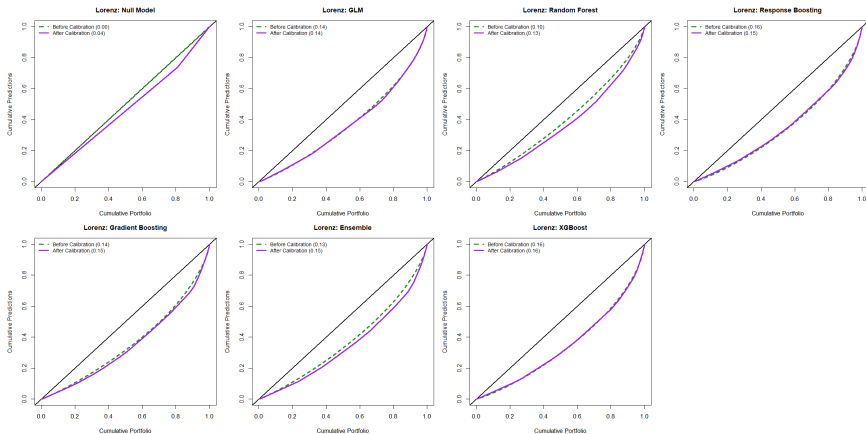
Concentration Curves

- For all analyzed models, the curves before calibration and after calibration almost perfectly overlap. The highest area under the curve metrics were recorded for the models: Gradient Boosting, Response Boosting, XGBoost, and Ensemble.



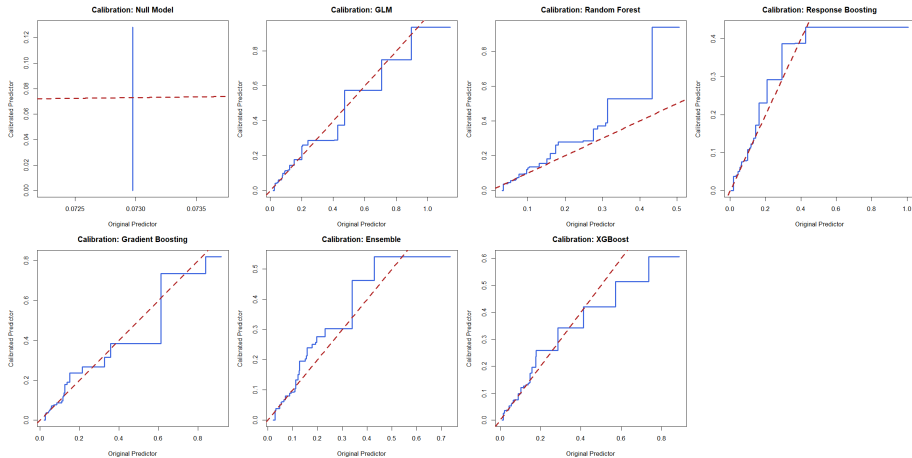
Lorenz Curves

- The Lorenz curve visualizes the model's ability to differentiate insurance risk, where the Null Model exhibits the lowest power, and the XGBoost and Response Boosting algorithms (0.16) exhibit the highest; the largest discrepancies in the curves before and after calibration occur in the Null Model and Random Forest, indicating a need for their correction.



Calibration Plots

- Calibration plots illustrate the relationship between model predictions and their values after the calibration process, where the Random Forest and the constant predictor of the Null Model require the greatest adjustments, whereas the GLM model demonstrates natural stability, oscillating closely to the optimal line.



Comparison of the Best Model with GLM

- Based on the conducted evaluation analysis, in which it dominated across almost all key metrics, XGBoost was selected as the best pricing model.
- In the area of high predictions, the green line of the average trend clearly falls below the $y = x$ axis. This indicates that for profiles with the highest claim frequency, XGBoost generates higher predictions than GLM, which explains its ability to discriminate the extreme segments of the portfolio.

